EXNO: 2	Analysis of SWR (Standing Wave Ratio)Transmission Lines by varying a)load impedance b) characteristic impedance
DATE:	

AIM: To analyze the SWR of a transmission line for a fixed characteristic impedance (Zo) of 50

ohms, as the load impedance (ZL) is varied across the following values: 25 ohms, 50 ohms, 75

ohms, and 100 ohms. To verify how the SWR changes with each value of ZL and when ZL =Zo.

SOFTWAREREQUIRED:

SCILABVersion6.1.1

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Typetheprogram
- Execute the program
- End the program

CALCULATION

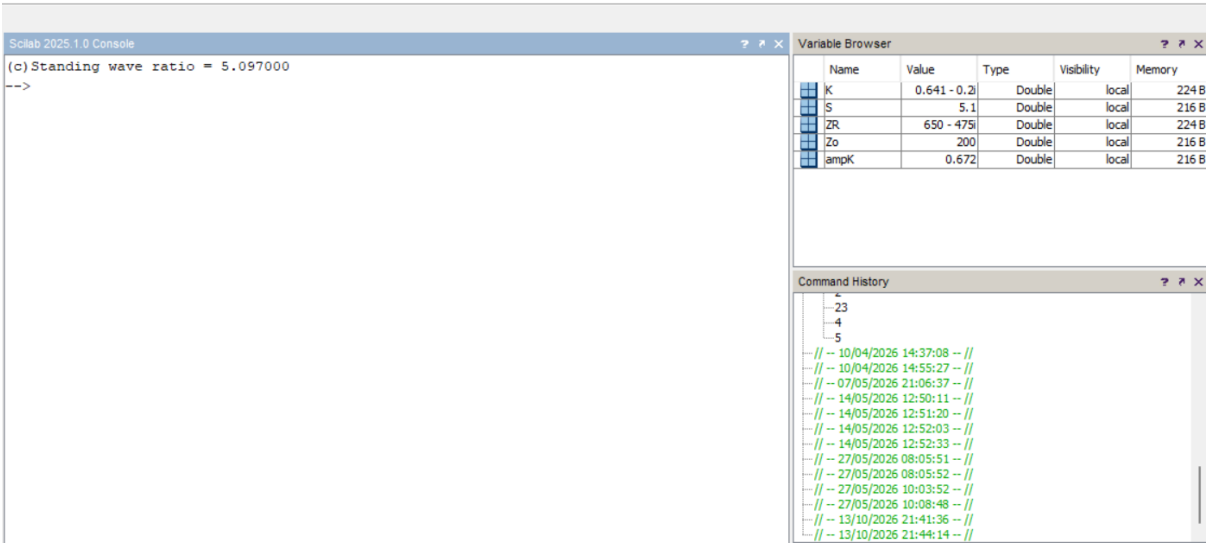
$$K = \frac{Z_R - R_0}{Z_R + R_0}$$

$$S = \frac{1 + |K|}{1 - |K|}$$

PROGRAM


SBEXP-2.sci (C:\Users\M.Pragadeesh\Documents\SBEXP-2.sci) - SciNotes
File Edit Format Options Window Execute ?
SBEXP-2.sci (C:\Users\M.Pragadeesh\Documents\SBEXP-2.sci) - SciNotes
SBExp-1.sci
SBEXP-2.sci
clc;
clear;
close;
Zo=200;
ZR=650-(%i*475);
K=(ZR-Zo)/(ZR+Zo);
ampK=sqrt((real(K)^2)+(imag(K)^2));
S=(1+ampK)/(1-ampK);
printf("(c)Standing wave ratio = %f",round(S*1000)/1000);

OUTPUT



RESULT:

The SWR of a transmission line for a fixed characteristic impedance (Zo) of 50 ohms, is analyzed for various values of load impedance ZL. SWR is also examined for the condition ZL.